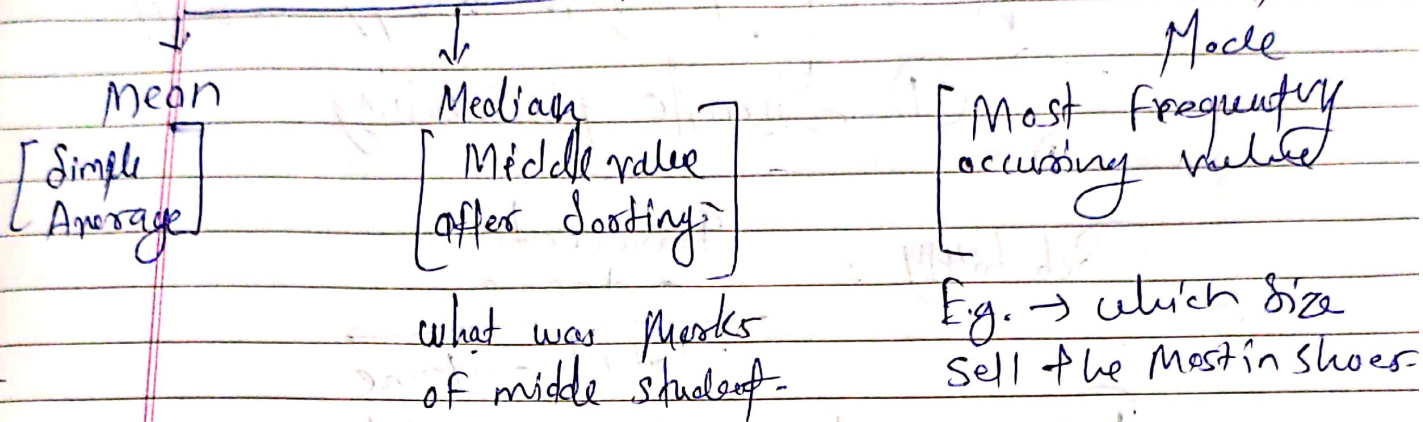


Central Tendency



SPREAD

$$\text{Variance} = \frac{1}{n} \sum (x_i - \bar{x})^2$$

mean

How far the data is from the average value

[Variance tells us if it is small then it is balance data all values are close]

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

why it exist because if we get variance of height 9 m^2 it does not make sense for unit so we make square root of it

$\sqrt{9} = 3 \text{ m} = \text{S.D.}$

Teacher's Signature:

~~Probability~~

PROBABILITY

① Simple Probability

$$\text{Probability} = \frac{\text{Favourite outcome}}{\text{Total outcome}}$$

$$\text{For coin} = \frac{1}{2} \quad \text{For Dice} = \frac{1}{6}$$

② Conditional Probability

In bag (R) (R) (R) (B) (B)
↓
3 Red 2 Blue
balls

$$P(R) = \frac{3}{5}$$

$$P(B) = \frac{2}{5}$$

After pick one Red ball
it become

$$P(R) = \frac{2}{4}$$

$$P(B) = \frac{2}{4}$$

So event is dependent

Formula

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

Discrete Values

[1, 2, 3, 4]

No. of students

Continuous Values

[1, 1.2, 1.3, 1.34, ...]

[time of delivery in M.S.]

Simulation

Toss a coin 100 times
Then give results.

Conditional Probability example.

Class

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

60 Boys

40 girls

45 play cricket.

(1) What is probability of a student plays cricket and give student is boy

A = Student play cricket

B = Student is boy.

$$P(B) = 60/100 = 0.6$$

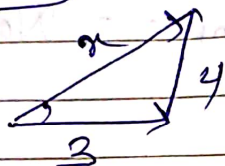
$$P(A \cap B) = 45/100 = 0.45$$

$$P(A/B) = \frac{0.45}{0.6} = 0.75$$

means student is both & Boy play cricket

LINEAR ALGEBRA

Vector



$$\theta = \tan^{-1} \left(\frac{\text{opp}}{\text{Adjacent}} \right)$$

Angle $\left[\theta = \tan^{-1} (4/3) = 53^\circ \right]$

$$r = \sqrt{4^2 + 3^2}$$

← Euclidean distance

$$r = 5$$

Magnitude

Matrix

Columns. ↓

Rows →

$$\begin{bmatrix} 5 & 3 & 8 \\ 7 & 2 & 6 \end{bmatrix}$$

Eg. in two stores they sold these items pen, pencil, eraser

Make matrix rows will define store column will be items

pen pencil eraser

Store A

Store B

$$\begin{bmatrix} 10 & 5 & 3 \\ 8 & 12 & 4 \end{bmatrix}$$

How to find position

$$M[R][C] = M[1][2] = 12$$

DATE: / /

Matrix Multiplication

Items ↓ Prices

Store → $\begin{bmatrix} 10 & 5 & 3 \\ 8 & 12 & 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$

For Matrix Multiplication
 len No. of Columns = No. of Rows Matrix B

$$10 \times 4 + 5 \times 2 + 3 \times 1 = \text{Total Sales Store A} = 53 \text{ Rs}$$

$$8 \times 4 + 12 \times 2 + 4 \times 1 = 32 + 24 + 4 = 60 \text{ Rs}$$

Dot products
Vector similarity
 Marks

Student 1 = $\begin{bmatrix} 80 & 90 & 70 \end{bmatrix}$

Student 2 = $\begin{bmatrix} 85 & 95 & 75 \end{bmatrix}$

Similarity Score = $(80 \times 85) + (90 \times 95) + (70 \times 75)$
 $= 20,600$

Teacher's Signature: